Weber (Yen-Chun) Huang

I am an experienced AI Engineer specializing in the training, deployment, and productionization of Large Language Models (LLMs). I bring a deep R&D background in both Natural Language Processing (NLP) and Computer Vision (image processing). Currently, I am actively exploring the frontier of AI Agent architectures and advanced LLM applications, and I am keen to translate these innovative technologies into practical products that deliver significant value.

Taipei, Taiwan

Work Experience

Sr Engineer • TrendMicro

Senior AI Engineer, Global IS Team | Driving Agentic AI Applications for Hyperautomation

- Architected and deployed a highly reliable, cross-regional intelligent customer service workflow using **LangGraph, FastAPI, and Azure Service Bus** to ensure operational scalability and seamless support resolution.
- Pioneered the **FastMCP** toolchain leveraging **FastAPI and Docker** to optimize content processing latency and markedly increase the intelligent bot's response accuracy via deep integration with the internal MCP-Server.
- Led collaboration with the D365 CRM Team to automate repetitive internal workflows, substantially reducing manual intervention time and significantly boosting overall team operational efficiency.
- Spearheaded R&D via POCs to improve bot accuracy and strategically control LLM Token consumption costs, demonstrating proactive resource management and cost-effectiveness.
- Championed internal technology adoption through technical sharing sessions to non-technical stakeholders, and initiated engineering workshops to provide strategic input on the future AI technology roadmap.

July 2025 - Present

Principal RD Engineer • TPIsoftware

Initiated and led the algorithm team in the development of **systalk.VIKI**, an internal enterprise generative AI assistant.

- Led the technical proposal for a two-stage Supervised Fine-Tuning (SFT) methodology for model parameters, serving as the company's technical lead for Large Language Model (LLM) finetuning.
- Collaborated with Product Managers and internal financial experts to develop asynchronous general financial knowledge datasets and instruction datasets using **Langchain**, **LangGraph**, and **Streamlit** via a Multi-LLM Agent collection platform.
- Directed senior engineers in building a model training platform based on **Huggingface** and **Pytorch**, incorporating data version control with **DVC (Data Version Control)** and model state tracking with **MLflow**.
- Fine-tuned **Breeze 7B** and **Llama 3.1** models using financial and regulatory data.
- Supervised technical interns in creating a training dataset management platform utilizing **Django Rest Framework**, **MongoDB**, and **OpenSearch**.
- Established an internal **Prompt Lab** and developed demo sites for various solutions (e.g., voice quality inspection, AI virtual human coaching), significantly improving the success rate of external AI pre-sales engagements and reducing AI product implementation costs for the data onboarding team.
- Initiated internal technical discussions and sharing sessions to foster cross-team AI knowledge exchange.
- Shared various prompt optimization strategies, including Chain-of-Thought (**CoT**), **ReAct**, and **Reflexion**.
- Encouraged team member initiatives, such as developing an internal Auto Prompt Optimization (**OPRO**) tool using **DsPy** and a graphical Agent Flow tool using **LangFlow**.
- Continuously researched the feasibility of adopting and integrating new technologies.
- Deployed and tested the performance of various open-source Chinese LLMs (e.g., **Breeze 8x7b**, **LLaMA**, **Gemma**) for enterprise and financial industry use cases using **Docker** and **vLLM**.
- Developed the **VIKI LLM Server** using **Django Rest Framework**, **OpenAI SDK**, and **PostgreSQL**. Collaborated with system developers to standardize internal model API interfaces.
- Established **VIKI** version control procedures (**Git flow** and **Docker** image management), optimizing the product deployment workflow.
- Assisted in the integration of **VIKI** into the company's internal HR system.
- Supported team efforts in acquiring external financial and organizational clients.
- Established team project management workflows using **GitHub Projects**, implemented a **Peer Code Review** system, and set up a documentation management platform, resulting in reduced product maintenance costs.

October 2024 - June 2025

Senior RD Engineer • TPIsoftware

Senior Algorithm Engineer at systalk.AI Algorithm Team

- Leveraged LLM and web API tools such as **Streamlit**, **Django Rest Framework**, **Langchain**, and **vLLM** to contribute to the planning and development of LLM products from the ground up, including the establishment of the company's primary product, **systalk.VIKI**.
- Developed crucial tools for the product team, including a **RAG (Retrieval-Augmented Generation)** knowledge base assistant and the **Systalk.LAB** prompt engineering validation tool, enhancing the team's capability for rapid validation of both cloud-based and on-premise model platforms.
- Developed an OCR model training and validation pipeline using **Pytorch**, based on the **Transformer architecture**, supporting the product team in building the foundation for **Systalk.ICR** (a product for intelligent recognition of diagnostic certificates, inpatient, and outpatient documents).
- Planned and built internal packages (**Systalk-ai-core**, **Systalk-nlp**, **Systalk-vision**) for data analysis, data cleaning, and machine learning, referencing Directed Acyclic Graph (**DAG**) algorithms and **Pytorch** pipelines, which successfully reduced development costs for ML training and validation code.
- Developed text mining APIs using **Django Rest Framework** to assist a life insurance client in optimizing their medical insurance form proofreading process.
- Optimized the existing conversational NLP customer service product, **Systalk.Chat**.

September 2022 - September 2024

NLP RD Engineer • eLand Information Co, Ltd

Work on the Natural Language Processing Team as NLP RD Engineer, responsible for the following:

- NLP machine learning and deep learning models tuning and development.
- Responsible for product planning, development and deployment of MLOps web API and user interface.
- User manuals and software documentation maintenance.
- Communicate and cooperate with internal users.
- Project CI/CD.

July 2021 - July 2022

Data Analyst • Raphael Clinic

- Assist in collecting and analyzing complex data.
- Create a project insight report.
- Provide strategic advice on marketing and digital activities to managers.
- Assist in the establishment of internal training materials for the company.

April 2019 - July 2019

Summer Intern • Taiwan Intellectual Property Office

The Information Services Group is to provide all colleagues in the Intellectual Property Office with knowledge and information related to business needs. Mainly responsible for content:

1. Organize exhibitions and activities to promote the information and documents of the Information Services Group to all colleagues in the Intellectual Property Office

2. Cataloging, filing and organizing new data

July 2017 - August 2017

Education

2019 - 2021

Taipei Medical University

Graduate Instute of Data Science

2014 - 2018

National Taiwan University

Library and Information Science

Skills

- Python / PyTorch
- Scikit Learn / OpenCV
- Docker & Docker Compose
- Git / GitLab
- Notion
- FastAPI / Django REST Framework
- Prompt / Context Engineer
- Langchain
- LangGraph
- LangFlow
- LLM Fine-tuning
- LLMops
- vLLM

Language

- English — Advanced level
- Chinese — Native language level
- Japanese — Intermediate level